

New Kochen–Specker sets from algebraic number fields: the Heegner-7 and golden ratio constructions

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We report two previously unknown Kochen–Specker (KS) sets in dimension 3 arising from algebraic number fields not previously explored in this context: the Heegner-7 ring $\mathbb{Z}[(1+\sqrt{-7})/2]$ yields a 43-vector set whose orthogonality graph is non-isomorphic to all known algebraic KS sets in dimension 3, and the golden ratio field $\mathbb{Q}(\sqrt{5})$ yields a 52-vector set not found by direct alphabet search and revealed only by Hermitian orthogonal completion. The Heegner-7 *full ray pool* (before minimization) has a graph-theoretic Lovász theta ratio $\vartheta/\alpha = 1.118$, the largest among the five algebraic KS ray pools with computed data. This ratio is driven by *auxiliary rays*—rays that have orthogonal neighbors but participate in no *designated-alphabet basis*—and collapses to $\vartheta = \alpha$ after cardinality minimization. The contrast identifies a structural tension between minimizing a KS proof and maximizing the graph-theoretic potential for contextual advantage.

INTRODUCTION

The Kochen–Specker (KS) theorem [1] proves that quantum observables cannot be assigned pre-existing values independently of measurement context. In dimension 3, all known KS constructions draw their ray coordinates from a handful of algebraic number fields: the integers [2], $\mathbb{Z}[\sqrt{2}]$ (the Peres construction [3]), and the Eisenstein integers $\mathbb{Z}[\omega]$ (Cabello’s recent simplest KS set [4]). Whether other number fields support KS sets, and what operational consequences follow, has not been systematically investigated. A central open problem is the minimum size of a KS set in dimension 3: the smallest known construction uses 31 vectors [2], while the best lower bound is 24 [6]. Closing this gap—either by finding a KS set with fewer than 31 rays or by proving that none exists—motivates a systematic exploration of the algebraic structures underlying KS constructions.

A survey across 40+ number fields reveals that only a handful support KS constructions, each determined by a specific cancellation identity in its ring of integers. None of the new constructions yield a KS set smaller than 31 vectors, but they expose structural features invisible in the integer construction alone. We report two new KS sets and identify a mechanism by which the algebraic origin controls the strength of the quantum contextual advantage in the Cabello–Severini–Winter (CSW) framework [5].

CONSTRUCTION METHOD

The classical KS constructions in $\mathbb{R}^3$ use rays whose coordinates are small integers: Conway and Kochen’s 31-vector set [2] uses directions (a, b, c) with $a, b, c \in \{0, \pm 1, \pm 2\}$ —geometrically, these point toward vertices, edge midpoints, and face centers of nested cubes, and the alphabet can be read directly from the coordinate labels on these cube diagrams. Orthogonality between two

such rays requires the dot product $\sum_k \bar{u}_k v_k = 0$, which for coordinates drawn from a finite set demands an algebraic relation—a *cancellation identity*—that makes the sum vanish exactly. The set of allowed coordinate values, which we call the *coordinate alphabet* $\mathcal{A}$ (following the “vector components” of Pavicic et al. [10]), thus determines which orthogonalities are available, and hence which KS sets can be constructed. For the integer alphabet, the key identity is $1 + 1 = 2$; for $\mathbb{Z}[\sqrt{2}]$ it is $(\sqrt{2})^2 = 2$; for the Eisenstein integers it is $1 + \omega + \omega^2 = 0$. Each identity enables exact orthogonal triples (bases), and when enough bases interlock, no consistent $\{0, 1\}$ -coloring exists—a KS set. Different algebraic number rings provide different alphabets with different cancellation identities. This paper’s approach is to construct new alphabets from algebraic number fields, identify their cancellation identities, and test whether the resulting ray sets are KS-uncolorable.

Concretely, given an alphabet $\mathcal{A} \subset \mathbb{C}$, we generate all rays $[v] \in \mathbb{CP}^2$ whose homogeneous coordinates lie in $\mathcal{A}$, identify orthogonal pairs via $\langle u|v \rangle = 0$, and collect orthonormal bases (triples of mutually orthogonal rays). KS-uncolorability is decided by SAT encoding: each ray i gets a Boolean variable x_i ; each basis $\{a, b, c\}$ contributes clauses enforcing $x_a + x_b + x_c = 1$; each orthogonal pair (i, j) contributes $\neg x_i \vee \neg x_j$. The encoding is solved by Glucose4 [7] via PySAT. We minimize by randomized greedy deletion (500 trials, random seed 42), which yields a small KS subset but does not guarantee minimality.

For fields where the basic alphabet yields too few bases, we apply *Hermitian orthogonal completion*. Given two orthogonal rays $u, v \in \mathbb{C}^3$ with $\langle u|v \rangle = 0$, their orthogonal complement $\text{span}\{u, v\}^\perp$ is one-dimensional. We compute its representative via the Hermitian cofactor formula:

$$w_k = (-1)^{k+1} \det \begin{pmatrix} \bar{u}_{k+1} & \bar{u}_{k+2} \\ \bar{v}_{k+1} & \bar{v}_{k+2} \end{pmatrix}, \quad k \in \mathbb{Z}/3\mathbb{Z},$$

and canonicalize w by fixing the first nonzero coordinate

to be real and positive. This is iterated: each new ray generates new orthogonal pairs, which generate further completions, until the pool stabilizes (an observed fixed point, typically reached in 2–4 rounds for the alphabets tested here; we do not prove termination in general). For real coordinates the formula reduces to the standard cross product. This closure can reveal KS sets invisible to the raw alphabet.

TWO NEW KS SETS

The Heegner-7 construction. The imaginary quadratic field $\mathbb{Q}(\sqrt{-7})$ has ring of integers $\mathbb{Z}[\alpha]$ where $\alpha = (1+\sqrt{-7})/2$ satisfies Hermitian squared modulus $|\alpha|^2 = \alpha\bar{\alpha} = 2$. (This field also has class number 1, though the property directly relevant to the construction is the existence of a modulus-2 generator.) The alphabet $\{0, \pm 1, \pm\alpha, \pm\bar{\alpha}\}$ generates 45 rays with 12 designated-alphabet bases. Hermitian orthogonal completion expands the ray pool to 145 rays and 42 designated bases. The completed pool is KS-uncolorable; randomized greedy deletion (500 trials) reduces it to **43 vectors in 23 bases**, the smallest KS subset found within this pool.

This 43-vector set has a distinct orthogonality graph: we verified non-isomorphism against the four other algebraic KS constructions known to us in dimension 3 (Conway–Kochen 31v, Peres 33v, Eisenstein 33v, $\mathbb{Z}[\sqrt{-2}]$ 33v) using degree sequence, spectral radius, $\text{tr}(A^4)$, and Weisfeiler–Leman 1-stable coloring (WL-1). The Heegner-7 graph differs from all four in degree sequence alone. A broader comparison against full KS catalogs (e.g. via canonical labeling with nauty/Traces) remains to be done. Explicit ray coordinates and the full orthogonality graph are provided as Supplemental Material [11].

The golden ratio construction. The golden ratio $\varphi = (1+\sqrt{5})/2$ satisfies $\varphi^2 = \varphi + 1$, giving Hermitian squared modulus $|\varphi|^2 \approx 2.618$. (Note: the algebraic norm $N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\varphi) = -1$ is distinct from $|\varphi|^2$; the relevant quantity for orthogonality is the Hermitian modulus.) The basic alphabet $\{0, \pm 1, \pm\varphi\}$ is colorable. However, Hermitian orthogonal completion generates rays with coordinates in the full ring $\mathbb{Z}[\varphi]$, including $\varphi - 1 = 1/\varphi$, and the completed pool of 205 rays is KS-uncolorable. Greedy deletion yields a smallest found subset of **52 vectors**.

This construction is not found by our direct alphabet search over $\{0, \pm 1, \pm\varphi\}$: the basic alphabet pool is colorable, and uncolorability emerges only after orthogonal completion introduces additional coordinate values.

Golden ratio coordinates have appeared in higher-dimensional KS catalogs (e.g., Pavicic et al. [10] in dimension 4), but no dimension-3 KS set from this field has been previously reported. To our knowledge, neither the Heegner-7 nor the dimension-3 golden ratio KS

TABLE I. CSW invariants for algebraic KS constructions. Top: minimized KS subsets. Bottom: full ray pools (complete closure before minimization). The ratio ϑ/α measures contextual advantage.

<i>Minimized KS sets</i>				
Construction	n	α	ϑ	ϑ/α
Integer (CK-31)	31	11	11.71	1.065
Eisenstein	33	12	12.05	1.004
Peres	33	12	12.00	1.000
$\mathbb{Z}[\sqrt{-2}]$	33	12	12.00	1.000
Heegner-7	43	16	16.00	1.000
<i>Full ray pools</i>				
Construction	n	α	ϑ	ϑ/α
Heegner-7	145	50	55.89	1.118
Eisenstein	57	18	19.34	1.074
Integer	49	17	17.70	1.041
Peres	49	23	23.00	1.000
$\mathbb{Z}[\sqrt{-2}]$	49	23	23.00	1.000

set has been previously reported. (For a comprehensive review of KS contextuality, see [12].)

CONTEXTUAL ADVANTAGE

The CSW framework [5] maps the orthogonality graph G of a ray set to a noncontextuality inequality via three invariants:

$$\alpha(G) \leq \vartheta(G) \leq \alpha^*(G),$$

where α is the independence number (classical bound), ϑ is the Lovász theta number (quantum bound), and α^* is the fractional packing number. A quantum advantage exists when $\vartheta > \alpha$.

Each ray v_i defines a rank-1 projective test $P_i = |v_i\rangle\langle v_i|$. Two rays with $\langle u|v\rangle = 0$ yield mutually exclusive outcomes—a consequence of Hilbert space structure. The exclusivity graph G encodes these constraints; $\alpha(G)$ bounds the noncontextual value of the corresponding CSW inequality, and $\vartheta(G)$ provides an upper bound on the quantum value optimized over all orthonormal representations and quantum states [5]. The quantum value actually realized by the *specific* projector set $\{P_i\}$ may be smaller than $\vartheta(G)$; we report ϑ/α as a graph-theoretic proxy for contextual potential, not as the achieved operational advantage.

Table I compares the CSW invariants for minimized KS sets and full ray pools across the known algebraic KS constructions. (The Eisenstein construction coincides with Cabello’s recent simplest KS set [4]; the golden ratio construction is omitted from the pool comparison because its 205-ray pool CSW invariants have not yet been computed.)

The Heegner-7 full pool has $\vartheta/\alpha = 1.118$, the largest graph-theoretic ratio among the five algebraic KS ray pools with computed CSW data. For comparison, CK-31’s minimized set has $\vartheta/\alpha = 1.065$, and most minimized KS sets are ϑ -perfect ($\vartheta = \alpha$). The Heegner-7 minimized 43-vector set itself satisfies $\vartheta(G) = \alpha(G)$ —the graph-theoretic gap is entirely a property of the larger pool.

THE AUXILIARY RAY MECHANISM

We identify the structural origin of this effect. In the Heegner-7 pool, “bases” are defined as orthonormal triples all of whose coordinates lie in the designated alphabet $\{0, \pm 1, \pm \alpha, \pm \bar{\alpha}\}$ and its closure. The pool is closed under Hermitian orthogonal completion: every orthogonal pair generates a third ray. However, the *coordinates* of that third ray may leave the designated alphabet, so the resulting triple is not counted as a basis for the KS coloring problem. This is a common convention in KS work: bases are the *designated contexts*, not all orthogonal triples that happen to exist in the pool.

A ray is *basis-participating* if it belongs to at least one designated basis, and *auxiliary* otherwise. Auxiliary rays have orthogonal neighbors—and hence edges in the orthogonality graph—but appear in no designated context.

KS-uncolorability depends on the basis *hypergraph*: auxiliary rays are invisible to it. The CSW framework depends on the orthogonality *graph*: auxiliary rays contribute edges that constrain the independence number α without adding bases. (That ϑ and α can change under vertex removal is a standard property of graph invariants; the contribution here is identifying *which* rays are dispensable for the KS hypergraph yet structurally important for the graph-theoretic bound.)

In the Heegner-7 pool, 76 of 145 rays (52%) are auxiliary. These rays add 522 orthogonal pairs, including 9 hub vertices of degree 16, severely restricting independent sets ($\alpha/n = 0.345$). The SDP relaxation exploits these constraints, distributing weight fractionally to achieve $\vartheta/n = 0.386$.

When minimization reduces the pool to 43 vectors, it removes auxiliary rays first—they are dispensable for uncolorability. But these are precisely the rays that enabled $\vartheta > \alpha$. The graph-theoretic gap is destroyed.

An ablation study confirms this: removing auxiliary rays in batches of 16, ϑ/α degrades smoothly from 1.118 through 1.09, 1.06, to 1.03, while KS-uncolorability and all 42 bases are preserved (since auxiliary rays do not participate in bases). Removing all 76 auxiliary rays collapses the gap entirely (full ablation data in Supplemental Material [11]).

Table II reports the Hoffman ratio H as a heuristic spectral indicator. Although H is not a valid bound for these irregular graphs, a suggestive pattern is visible across the five pools: pools with $\alpha/H < 1$ have

TABLE II. Auxiliary ray fraction and Hoffman ratio for the five pools with computed CSW data. $H = -n\lambda_{\min}/(\lambda_{\max} - \lambda_{\min})$ is the Hoffman ratio bound, which is a valid upper bound on α only for regular graphs. These orthogonality graphs are irregular, so H is used here as a heuristic spectral indicator, not a rigorous bound.

Construction	Aux. frac.	α/H	ϑ/α
Heegner-7	52%	0.977	1.118
Eisenstein	0%	0.904	1.074
Integer	0%	0.977	1.041
Peres	33%	1.176	1.000
$\mathbb{Z}[\sqrt{-2}]$	33%	1.176	1.000

$\vartheta > \alpha$, while those with $\alpha/H > 1$ satisfy $\vartheta = \alpha$. We caution that five data points are too few to establish a correlation; the pattern is reported as an observation, not a claim. The Eisenstein pool has no auxiliary rays yet still has $\vartheta/\alpha = 1.074$: here the gap is purely spectral, arising from edge density among basis-participating rays alone. The Peres and $\mathbb{Z}[\sqrt{-2}]$ pools have 33% auxiliary rays but no gap because their orthogonality graphs decompose into disconnected components; since ϑ and α are additive over disjoint unions [8], isolated auxiliary components cannot create a gap. The Heegner-7 pool combines both mechanisms: auxiliary rays that bridge into the basis-participating core, enabling the SDP to exploit their constraints. We compare only among the five algebraically generated KS ray pools known to us in dimension 3; we do not claim optimality over all projector configurations in $\mathbb{C}^3$.

DISCUSSION

Our results reveal a structural tension: *minimizing the proof size and maximizing the graph-theoretic contextual potential are competing objectives*. The rays that are dispensable for the logical proof of contextuality (the auxiliary rays) are precisely those that increase ϑ/α in the CSW framework. This suggests that the standard practice of seeking minimal KS sets, while natural from a proof-complexity perspective, discards structure relevant to the graph-theoretic potential for contextual advantage.

The Heegner-7 pool’s 12% ϑ/α ratio is modest but structurally informative: it arises from the interplay between the number field (which determines auxiliary ray density) and the spectral properties of the orthogonality graph (which determine whether auxiliary rays translate into a $\vartheta > \alpha$ gap). Whether this graph-theoretic gap is realized operationally by the specific projector set remains to be determined. The golden ratio construction demonstrates that geometric closure can unlock contextuality where algebra alone cannot.

KS sets also have direct operational significance beyond CSW inequalities: Cabello [9] proved that ev-

ery bipartite perfect quantum strategy requires a KS-uncolorable subset, so the Heegner-7 (23 bases) and golden ratio (25 bases) sets are candidates for new bipartite Bell scenarios that may admit perfect quantum strategies.

Both constructions arise from a pattern observed across all tested number fields: KS-uncolorability appears to require one of two cancellation mechanisms—*modulus-2 cancellation* (a generator x with Hermitian squared modulus $|x|^2 = 2$, enabling identities such as $1+1 = 2$, $(\sqrt{2})^2 = 2$, $\alpha\bar{\alpha} = 2$) or *phase cancellation* (a root-of-unity sum vanishing exactly, as in $1+\omega+\omega^2 = 0$). In all fields we tested, alphabets whose smallest available cancellation involves $|x|^2 \geq 3$ produce orthogonal triples but not KS-uncolorability. The golden ratio case ($|\varphi|^2 \approx 2.618$) does not violate this pattern: cross-product completion introduces modulus-2-type cancellations into the effective coordinate set.

For the restricted family of two-element alphabets $\{0, \pm 1, \pm x\}$, we can enumerate all non-trivial three-term zero-sums from the product set $\{0, \pm 1, \pm x, \pm |x|^2\}$: the only solutions are $x = 2$ (integer), $|x|^2 = 2$ (Peres, $\mathbb{Z}[\sqrt{-2}]$, Heegner-7), $|x|^2 = x + 1$ (golden ratio), and $x^2 + x + 1 = 0$ (Eisenstein). This enumeration is complete for two-element alphabets, but multi-element alphabets from higher-degree number fields remain unexplored: cross-products xy may introduce cancellation identities not reducible to these four types.

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